This idea, known as energy harvesting, has caught many people's interest. Even though there are limitations for the large-scale applications, you can produce electricity that is enough to charge your mobile phones by just walking. There are power generating footwears that has a slip-on insole with piezoelectric crystals that can produce enough electricity to charge batteries/ USB devices.`

Summary

- Solids have definite volume and shape.
- Solids can be classified into the following two major types based on the arrangement of their constituents. (i) Crystalline solids (ii)Amorphous solids.
- A crystalline solid is one in which its constituents (atoms, ions or molecules), have an orderly arrangement extending over a long range.
- In contrast, in amorphous solids (In Greek, amorphous means no form) the constituents are randomly arranged.
- Crystalline solid is characterised by a definite orientation of atoms, ions or molecules, relative to one another in a three dimensional pattern. The regular arrangement of these species throughout the crystal is called a crystal lattice.
- A crystal may be considered to consist of large number of unit cells, each one in direct contact with its nearer neighbour and all similarly oriented in space.
- A unit cell is characterised by the three edge lengths or lattice constants a ,b and c and the angle between the edges α, β and γ
- There are seven primitive crystal systems; cubic, tetragonal, orthorhombic, hexagonal, monoclinic, triclinic and rhombohedral. They differ in the arrangement of their crystallographic axes and angles. Corresponding to the above seven, Bravis defined 14 possible crystal systems
- In the simple cubic unit cell, each corner is occupied by an identical atoms or ions or molecules. And they touch along the edges of the cube, do not touch diagonally. The coordination number of each atom is 6.
- In a body centered cubic unit cell, each corner is occupied by an identical particle and in addition to that one atom occupies the body centre. Those atoms which occupy the corners do not touch each other, however they all touch the one that occupies the body centre. Hence, each atom is surrounded by eight nearest neighbours and coordination number is 8.
- In a face centered cubic unit cell, identical atoms lie at each corner as well as in the centre of each face. Those atoms in the corners touch those in the faces but not each other.The coordination number is 12.

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